

NEXT GENERATION ELEVATOR CHALLENGE

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Q1: IF ONE BUILDING HAS FOUR ELEVATORS, AND ONE PERSON CALLS THESE ALL BECAUSE HE IS URGENT. AT THAT TIME, IF ALL ELEVATORS ARE COMING, ENERGY USAGE IS WASTE? HOW TO SOLVE IT?.

- In control terms, current.
 - Use a central group controller with one-call-one-car assignment

Example cost idea:

```
cost =  
    distance_weight * floors_away  
+ stop_weight * number_of_stops_before_arrival  
+ reverse_penalty if direction_must_change  
+ load_penalty if car_is_near_full  
+ energy_weight * estimated_energy
```

Q2: WHEN ELEVATOR IS GOING UP, ONE WOMAN WANTS TO GO DOWN. BUT IF SHE ENTERS THAT ELEVATOR, SHE HAS TO GO UP FIRST AND THEN GO DOWN AGAIN. SHE DOESN'T WANT TO GO UP. HOW TO SOLVE IT?

- In HCI terms, the weakness is poor feedback.
 - Show sign of Next direction after stop
 - Predefined audio announcement: “Please wait for the next down car. This elevator is going up.”
- In control terms, the weakness is bad assignment.
 - First choice: elevator already moving down and approaching
 - Second choice: idle nearby elevator
 - Last choice: elevator moving up that would force reverse or extra ride

Q3: PREVENT A PASSENGER FROM CALLING OR ENTERING AN ELEVATOR THAT IS UNDER REPAIR, AND PREVENT UNSAFE MOVEMENT WHILE A TECHNICIAN IS WORKING.

- In HCI terms, the weakness is poor feedback.
 - Show clear warnings sign 'Out of Service'
- In control terms, the weakness is bad assignment.
 - Hide it from dispatch
 - Block unsafe movement

Thank you